

QUANTICO MCAF, VA, USA

WMO: 724035

Lat: 38.504N

Lon: 77.305W

Elev: 10

StdP: 14.69

Time Zone: -5.00 (NAE)

Period: 94-19

WBAN: 13773

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) 15.9	(c) 19.6	(d) -2.4	(e) 4.8	(f) 20.9	(g) 2.3	(h) 6.2	(i) 24.2	(j) 25.3	(k) 36.5	(l) 22.6	(m) 38.1	(n) 8.0	(o) 290	(p) 0.457

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	16.3	92.5	76.3	90.2	75.8	87.9	75.0	79.8	88.3	78.2	86.6	77.0	84.9	7.7	190

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
77.3	142.3	84.9	75.7	134.5	83.0	74.5	129.3	81.8	43.2	88.1	41.6	86.9	40.3	85.0	90.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a) 20.7	(b) 18.2	(c) 16.2	(e) 8.9	(f) 97.2	(g) 6.0	(h) 3.3	(i) 4.6	(j) 99.6	(k) 1.1	(l) 101.5	(m) -2.3	(n) 103.3	(o) -6.6	(p) 105.7		
(4) 20.7	(5) 18.2	(6) 16.2	(7) 8.9	(8) 97.2	(9) 6.0	(10) 3.3	(11) 4.6	(12) 99.6	(13) 1.1	(14) 101.5	(15) -2.3	(16) 103.3	(17) -6.6	(18) 105.7		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	57.7	35.7	38.6	46.3	57.2	65.8	74.4	78.6	77.1	70.6	58.9	47.9	40.0
	DBStd	16.81	9.57	9.06	9.23	8.04	7.37	5.61	4.73	4.66	6.20	7.66	7.86	8.24
	HDD50	1468	453	334	186	22	0	0	0	0	12	131	330	
	HDD65	4109	908	739	584	259	84	5	0	19	219	515	777	
	CDD50	4274	11	14	70	238	490	733	886	839	617	289	68	19
	CDD65	1436	0	0	3	24	108	288	421	374	186	31	1	0
	CDH74	12005	1	2	29	182	714	2351	4055	3285	1232	147	6	1
	CDH80	4046	0	0	4	42	170	766	1587	1142	313	22	0	0
	WSAvg	7.3	7.5	7.6	8.7	8.5	7.3	6.8	6.4	6.5	7.1	7.1	7.3	7.0
	PrecAvg	38.1	2.9	3.0	3.4	2.8	3.6	3.2	3.2	3.7	3.6	2.6	3.2	3.1
(15) Precipitation	PrecMax	52.6	5.7	8.2	6.1	5.7	9.2	10.2	7.2	8.5	13.5	6.5	8.0	7.8
	PrecMin	21.4	1.3	0.4	1.5	0.2	0.6	0.7	0.8	0.5	0.1	0.0	0.4	0.2
	PrecStd	8.2	0.9	1.9	1.3	1.4	2.1	2.4	1.6	2.1	3.0	1.6	2.5	1.8
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4% DB	66.0	70.0	78.5	85.4	88.9	93.6	97.4	94.6	91.2	82.3	73.4	68.1	
	0.4% MCWB	57.2	58.3	60.4	64.9	72.1	75.5	78.3	77.3	75.2	72.0	61.1	59.1	
	2% DB	59.3	62.1	71.0	79.3	84.3	90.0	93.2	91.4	86.7	78.0	67.6	61.5	
	2% MCWB	52.4	53.0	57.1	63.3	71.0	74.8	77.1	76.6	74.1	68.9	58.5	55.1	
	5% DB	53.7	56.1	65.8	74.5	81.3	87.2	90.4	89.1	83.4	74.4	63.6	56.3	
	5% MCWB	48.7	47.9	55.2	61.9	69.9	74.3	76.6	75.8	72.4	67.0	56.5	51.2	
	10% DB	48.9	51.9	61.0	70.5	78.2	84.4	88.1	86.3	80.8	71.6	60.6	52.2	
	10% MCWB	44.2	45.3	51.5	59.9	67.6	73.0	75.6	74.5	71.0	64.7	54.4	47.4	
	0.4% WB	58.8	60.5	63.5	70.4	76.4	80.1	82.5	81.9	78.8	75.0	65.3	61.8	
	0.4% MCDB	64.6	66.6	72.4	76.5	84.1	88.5	90.6	88.9	85.4	79.1	68.7	65.9	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2% WB	54.0	54.5	59.8	66.4	73.6	77.9	80.1	79.5	76.5	72.1	61.7	56.8	
	2% MCDB	57.6	59.8	68.8	74.6	81.1	86.3	88.9	87.6	83.2	76.0	65.6	60.6	
	5% WB	49.4	49.8	56.6	64.0	71.7	76.4	78.5	77.8	74.9	69.0	58.5	52.6	
	5% MCDB	52.9	54.5	64.2	71.8	78.7	84.2	87.4	85.8	81.1	73.2	62.5	55.6	
	10% WB	45.0	45.7	52.7	61.6	69.7	75.0	77.2	76.4	73.3	65.6	55.4	47.9	
	10% MCDB	48.2	50.9	59.4	69.5	76.1	82.1	85.6	83.6	78.3	70.1	59.1	51.2	
(35) Mean Daily Temperature Range	MDBR	15.1	16.7	17.9	19.3	17.4	16.6	16.3	16.1	16.3	17.7	17.4	15.2	
	5% DB MCDBR	22.0	23.9	25.0	24.4	20.6	19.2	18.6	18.4	18.6	20.1	21.7	21.1	
	5% DB MCWBR	16.6	16.5	14.6	12.9	11.4	9.9	8.6	8.3	9.3	12.2	14.9	16.4	
	5% WB MCDBR	20.0	21.7	22.6	21.0	17.5	17.0	17.1	16.4	15.4	16.4	18.0	19.3	
	5% WB MCWBR	16.7	16.8	15.2	13.2	10.3	9.5	8.8	8.5	8.8	11.1	14.4	16.4	
(40) Clear-Sky Solar Irradiance	taub	0.306	0.321	0.356	0.401	0.440	0.497	0.518	0.515	0.425	0.360	0.324	0.310	
	taud	2.502	2.442	2.401	2.292	2.214	2.093	2.058	2.045	2.278	2.460	2.519	2.528	
	Ebn at Noon	278	287	287	278	268	252	246	242	260	270	269	268	
	Edh at Noon	25	30	35	41	45	51	52	52	39	29	24	23	
(44) All-Sky Solar Radiation	RadAvg	682	942	1255	1565	1773	1951	1901	1702	1391	1057	772	601	
	RadStd	59	103	90	114	189	147	97	85	160	118	64	57	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only		+1.21	N/A	N/A	N/A	N/A	N/A	N/A	+295	+205
(47) Regional (74 neighbors)		+0.79	N/A	N/A	N/A	N/A	N/A	N/A	+239	+142

Nomenclature: See separate page